Session Objectives

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Discriminate between the clinical presentations of typical viral exanthems based on exam findings and associated symptoms.
- Understand complications of typical viral exanthems and populations at higher risk.
- Discriminate between the diagnostic, treatment and prevention measures of typical viral exanthems.

Clinical Considerations

- Most are benign
- Can be devastating for certain populations
  - Immunocompromised, unvaccinated, fetus/newborn
- Treatment is generally supportive, but targeted anti-viral treatment is available for some
- Diagnosis is primarily clinical, but specific testing is available in certain circumstances

Viral Exanthems

- Rubella
- Roseola
- Erythema Infectiosum
- Measles
- Chicken Pox (Varicella)
- Hand, Foot, and Mouth Disease
- Herpes Simplex
- Pityriasis rosea
- Molluscum contagiousum
- Monkeypox

Case #1

- A 2 yo male was recently diagnosed with a viral illness by his pediatrician after presenting with fatigue, runny nose, fever and rash. The rash started on the cheeks and then spread to trunk and extremities.
- Today, his 15 yo sibling, who has sickle cell anemia, presents with increased pallor and fatigue. Exam is notable for tachycardia and a systolic ejection murmur.
- Mom has a picture of the toddler’s rash.

Erythema Infectiosum (5th Disease)

- Prodrome: mild “flu-like” illness [2-5 days]
  - Fever, malaise, myalgia, headache, coryza, diarrhea, abdominal pain
- Exanthem
  - Macular erythema of cheeks (“Slapped cheeks”) [2-4 days]
    - Sparing nasal, perioral, and periorbital areas
  - Facial rash fades and a pruritic reticular rash appears symmetrically on trunk and extremities [duration varies]
- No longer contagious once rash appears
- Treatment: supportive
- Diagnosis: clinical

Parvovirus B19: Other Clinical Presentations

- Symmetric small joint polyarthropathy
  - More common in adults
- Transient aplastic crisis
  - Occurs in patients with hemoglobinopathies (sickle cell), hemolytic anemias
- Chronic anemia
  - Occurs in immunosuppressed or HIV infected patients
- Hydrops fetalis
- Papular-purpuric gloves-and-socks syndrome (PPGSS)

Papular-purpuric gloves-and-socks syndrome (PPGSS)

- Adolescent girls, young adults
- Itching, burning, bright red papular exanthem on hands and feet
- Sharp border
- Edema can limit movement
- Can also see petechiae, purpura, oral vesicles
- Can also have fever, lymphadenopathy, arthralgias

Gloves and socks syndrome secondary to parvovirus B19 infection

Papular-purpuric rash on the hands and feet of a patient with acute parvovirus B19 infection.
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Case #2

- A 9 mo male presents to your office with fever of 102.5°F for 3 days. He has decreased appetite and increased fussiness. Exam is notable for mild cervical LAD. No treatment is recommended other than supportive care.
- The mother calls the next day and tells you the fever is gone but her son now has a pink rash all over his body. He is otherwise doing well (eating, drinking, playful).

Roseola

- High fever followed by a maculopapular rash that presents as the fever resolves
- Begins on trunk and spreads to extremities/neck/face - centrifugal
- Treatment: supportive
- Diagnosis: clinical
- Transmission: direct contact with saliva

HHV6: Other Clinical Presentations

- Can remain latent and reactivate later in life
- Can cause meningitis, encephalitis in immunocompromised patients
- In patients who have undergone hematopoietic stem cell transplant ganciclovir and foscarnet may be beneficial
- Testing in these more extreme cases should be done with PCR
  - Blood, CSF, tissue

Congenital Rubella Syndrome

- Cataracts, retinopathy, glaucoma
- Sensorineural hearing loss
- Patent ductus arteriosus, pulmonic stenosis
- Hepatosplenomegaly, jaundice
- Growth restriction
- Intellectual disability
- Erythematous maculopapular rash
    - Dermal erythropoiesis “Blueberry muffin”
- Diagnosis: serology (IgM, IgG), PCR (throat, nasal, urine), viral cx (throat, nasal, urine)

Rubella (German Measles)

- Prodrome: mild or absent
  - Low grade fever, sore throat, malaise
- Lymphadenopathy: posterior auricular, occipital, can be generalized
- Exanthem: fine, erythematous, maculopapular rash [1-3 days]
  - Starts on face and becomes generalized in 24 hours,
  - Enanthem: petechiae on soft palate (Forscheimer spots)
- Transmission: droplet, direct contact with nasopharyngeal secretions, transplacental
- Peaks in late winter/early spring
- Most infections now occur outside US or in under-immunized individuals
- Treatment: supportive care

Measles

- Incubation: asymptomatic [10-14 days]
- Prodrome: high grade fever, cough, coryza, conjunctivitis, pharyngitis, enanthem (Koplik spots) [2-3 days]
- Exanthem: starts 1-2 days after prodrome
  - Erythematous nonpruritic macules/patches on face
    - Koplik spots start to resolve prior to facial rash
  - Spreads to the trunk and extremities within 2-3 days
  - Palms and soles involved

Measles

- Diagnosis: clinical but can confirm with testing
  - Immunofluorescence of nasal mucosal cells
  - PCR of nasal secretions, blood, urine
  - Serology: IgM, 4 fold increase in IgG
  - Viral cell culture
- Treatment: vitamin A, ribavirin
- Postexposure prophylaxis: IVIG or vaccination with MMR/MMRV within 72 hours of exposure

Measles Complications

- Diarrhea
- Otitis media
- Croup, bronchiolitis
- Pneumonia is the most common cause of death from measles in young children
  - Primary viral or bacterial superinfection
- Encephalitis (15% mortality, 20-40% long-term sequelae)
  - Post-infectious, immune mediated
  - Uncommon – 1 in 1,000 cases
- Myocarditis
- Subacute sclerosing panencephalitis (SSPE)
  - Late complication, rare
  - Virus remains latent in CNS and reactivates 7-10 years later
  - Attacks CNS cells, leads to cell death, neurodegenerative process

Hand, Foot and Mouth Disease

- Caused by enteroviruses: coxsackie viruses, enteroviruses, echoviruses
- Peaks in summer/early fall
- Transmission: fecal-oral, respiratory
- Most commonly seen in children <10 yo
- Diagnosis: clinical but can confirm with PCR (vesicle fluid, blood, urine, CSF, stool, respiratory secretions)
- Treatment: supportive

Hand, Foot and Mouth Disease

- Prodrome: absent or mild [4-6 days]
  - Vomiting, diarrhea, abdominal pain
- Low grade fever can precede enanthem and exanthem
- Enanthem: painful, erythematous macules and vesicles
  - Commonly rupture and form superficial ulcer with gray-yellow base and erythematous rim
  - Refusal to eat or decreased oral intake is common
  - If oral involvement only it is referred to as herpangina
- Exanthem: macules, papules, cloudy vesicles with erythematous base (halo), non pruritic, usually nontender
  - Occurs on the hands, palms, feet, soles, buttocks, extremities

Case #6

- A 6 yo male presents with fever, cold symptoms, headache, fatigue and diffuse pruritic rash. The rash started 2 days after the fever. It was initially on the trunk and spread to all extremities. Some lesions are now crusted over. There are also some painful lesions in the mouth. He is with foster parents. They are not sure what immunizations the child has received.

Chicken Pox (Varicella)

- Prodrome: fever, fatigue, anorexia, congestion, rhinorrhea, cough
- Exanthem: very pruritic erythematous rash on trunk and scalp and then spreads peripherally (centrifugal)
  - Macules > papules > vesicles > pustules with crusting
  - Lesions at various stages, “Dew drop on rose petal”
- Enanthem: uncommon, painful vesicles > erosions on palate
- Transmission: airborne, direct contact of lesions, transplacental
- Diagnosis: clinical
  - Testing with PCR, DFA, culture if needed
- Treatment: supportive

Chicken Pox Complications

- Mainly in the immunocompromised
- Treat with antivirals in these patients: acyclovir, valacyclovir, famciclovir
- Bacterial superinfection (staph, strep): impetigo, sepsis, osteomyelitis
- Pneumonia: primary viral or bacterial superinfection
- Neuro: meningitis, encephalitis, Guillain-Barre Syndrome
- Reye syndrome

Case #7

- An 18 yo male with history of asthma just completed a course of prednisone for treatment of a severe asthma exacerbation. He presents to your office today for follow up and states that he is having “chest pain”. He has no respiratory or other concerning symptoms. On exam you note the following:

Shingles (Herpes zoster)

- Prodrome: Pain, tenderness, paresthesia, allodynia, intense itching in affected dermatome often precedes rash
- Exanthem: erythematous papules [24 hrs] > vesicles-bullae [48 hrs] > pustules [96 hrs] > crusts [7 to 10 days]
  - New lesions continue to appear for up to 1 week
  - Can affect adjacent dermatome (e.g. T4 and T5)
- Enanthem: painful erythematous vesicles and erosions can appear on mucous membranes of affected dermatome (i.e. mouth, vagina, urethra)
- Complications: postherpetic neuralgia occurs in ~40% of pts >60 yo
  - Resolution in 87% of pts at 6 mos

Ramsey Hunt Syndrome (Herpes Zoster Oticus)

- Involvement of geniculate ganglion and CNVIII
- Prodrome: Triad of 1) ipsilateral facial paralysis 2) ear pain 3) vesicles in auditory canal and auricle
- Symptoms in addition to exanthem
  - Ipsilateral altered taste, decreased hearing, tinnitus, lacrimation, vertigo
- Treatment: acyclovir, valacyclovir, famciclovir and steroids

Herpes Zoster Ophthalmicus

- Involvement of ophthalmic division of CNV
- Prodrome: headache, malaise, fever
- Symptoms in addition to exanthem
  - Unilateral pain of eye, forehead and top of head
  - Conjunctivitis, uveitis, episcleritis, keratitis
- Complications: potential loss of vision (early diagnosis critical)
- Treatment: acyclovir, valacyclovir, famciclovir

Herpes Simplex Virus (1 and 2)

- Typically, HSV1 causes cold sores and HSV2 causes genital herpes, **BUT**, both viruses can cause oral or genital infection (Xu J Inf Disease 2002)
- Each virus causes more severe disease if infecting it's typical location
- HSV1 prevalence: more than 25% of children will be infected by the age of 7 and more than 30% of adolescents will be infected by 19 yo (Xu J Pediatr 2007)
- Primary infection will have more systemic symptoms than recurrent infections
- Transmission: saliva, skin-skin contact through vesicular fluid, sexual

Herpes Simplex Virus Manifestations

- Oral: gingivostomatitis, labialis, pharyngitis
- Skin
  - Herpetic whitlow: infection of the finger
    - In children, can occur at the time of primary oral infection through autoinoculation
  - Herpetic gladitorum: infection of face, neck and arms; most commonly in wrestlers
  - Eczema herpeticum: hx of atopic dermatitis where HSV causes a secondary infection
  - Erythema multiforme
- Ocular: keratoconjunctivitis, chorioretinitis (neonates)
- Diagnosis: clinical or obtain swab for PCR from the vesicular fluid or ulcer

HSV Recurrence

- Reactivation triggers: sun, menses, trauma, decreased immunity
- Tend to be milder than initial presentation in immunocompetent
- Can be severe in immunosuppressed: disseminated disease, encephalitis, meningitis, keratitis
- Treatment: supportive, acyclovir, valacyclovir, famciclovir as needed
  - IV antivirals for neonates, encephalitis, immunocompromised
  - Can use antivirals daily for prophylaxis to reduce severity, frequency, duration of recurrent infections (HSV1 recurrence typically less than HSV2 recurrence; 1x/yr vs 4x/yr on avg)

Case #9

- An 18 yo male presents with chief complaint of “itchy rash”. Rash started one week ago with a single spot but has spread to his back, abdomen and chest. He has no other symptoms. He takes no medications. He has not spent a significant amount of time outside recently.

Pityriasis Rosea

- Exanthem:
  - Herald patch: 3-5cm isolated, oval, scaly pink patch with central clearing
  - Over 1-2 weeks will develop very pruritic, scaly, erythematous plaques, with colarette of scale in Christmas tree pattern (spreads cephalo-caudal)
- Diagnosis: clinical
  - Testing to exclude tinea or syphilis may be considered
- Treatment: supportive (antihistamine, topical steroid, UV light)
- Will resolve spontaneously in 6-8 wks

Molluscum

- Exanthem: discrete flesh-colored domed papules with central umbilication (2-5 mm)
  - Common on trunk, extremities, face
  - Asymptomatic generally but may be pruritic
  - Can get local eczematous reaction
- Transmission: skin to skin contact, autoinoculation through scratching
- Treatment: curettage, topicals, light therapy (cosmetic)
- Commonly resolves spontaneously over 2-24 months

Case #11

- A 7-year-old boy is brought to the pediatric clinic by his mother with a 3-day history of fever, fatigue, and a spreading rash. The rash initially appeared as small red spots on his face but has since progressed to involve his arms, legs, and trunk. Exam reveals erythematous pustules. Mom reports the child is up to date on vaccinations including varicella. The family recently returned from a Mediterranean cruise.

Monkeypox (Mpox)

- Viral zoonotic infection caused by monkeypox virus (MPXV)
- Transmission: skin-to-skin or mucosal contact (fomite possibly)
- Prodrome: fever, chills, myalgias, malaise [1-5 days]
- Exanthem: develops 1-3 days after fever and starts on the face and extremities (including palms and soles) with centripetal spread
    - Macules [1-2 days] > papules [1-2 days] > vesicles [1-2 days] > pustules [5-7 days] > crusts [up to 14 days]
    - Lesions in any one area of the body will be at the same stage and progress simultaneously

Mpox Treatment and Vaccine

- Treatment: supportive, most recover without any medical intervention.
    - Tecoviromat: originally developed to treat smallpox, used during outbreak under an Expanded Access Investigational New Drug protocol in the US
- Two vaccines available that may reduce risk of developing Mpox
    - MVA: made from a highly attenuated, nonreplicating vaccinia virus.
        - Approved in US for smallpox and monkeypox
    - ACAM2000: replication-competent smallpox vaccine
        - Only used in select patients and is associated with more adverse events
        - Only approved in US for the prevention of smallpox
        - Can be used for the prevention of monkeypox under an expanded-access investigational new drug (IND)

Summary Slide

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- Specific items in the history are very important for a patient presenting with a rash – timing, evolution, spread
- Describe the rash using terminology of primary and secondary lesions
- Viral exanthems can have similar appearances, but the history, distribution, and evolution of the rash can help with the differential diagnosis
- Viral exanthems can often be diagnosed clinically, but confirmatory testing is used in certain situations
- Treatment for viral exanthems is often supportive, but some antiviral agents are available and are used in special situations such as in immunocompromised patients
- While many viral exanthems are self-limited and mild infections, certain populations are at risk for more severe infection and complications